

10.2 Editing and Unlearning

ROME/MEMIT locate a fact and rewrite it. Unlearning tries to forget a set, not one key.

These notes match the lecture slides. Use **(slides)** on the course hub for the deck.

Weights store facts as distributed circuitry. **Editing** tries to change **one** association. **Unlearning** tries to remove a **set** of examples or a topic so they are no longer easy to recover. Both are approximate; neither is a legal delete button.

1. ROME: locate, then write

Meng et al., **ROME** (Rank-One Model Editing). Empirically, many “subject → attribute” facts (Paris, capital of, France) concentrate in a mid-layer MLP on the last token of the subject. **Locate** that layer with causal traces; **write** a rank-one update so the same key maps to a new value.

Locate, then write. One key, not a new pretraining run.



MEMIT extends the write across several layers for many facts at once. The algebra is a constrained least-squares update of the MLP down-projection, not a full fine-tune. Side effects: neighborhood facts can move; paraphrases may still leak the old value.

2. Unlearn a set versus edit a key

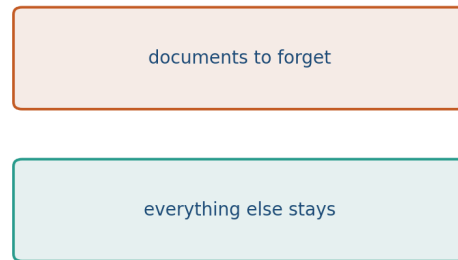
Edit a key: one prompt, one new completion (The capital of France is Berlin as a toy). Success is that prompt (and close paraphrases) flipping, while unrelated geography stays put.

Unlearn a set: a corpus you want gone (a book, personal data, a toxic cluster). Methods: gradient ascent on those rows, influence-style downweighting, relabel-and-SFT, or extra DPO against the bad completions. You must check **utility** on a retain set, not only forget accuracy. Models memorize by many routes; one loss spike does not erase a concept.

Edit one association



Unlearn a set



Lab 10 will overwrite a dictionary key—that is editing as data structure, not as ROME. The analogy: a key–value store is parametric memory you can assign; an LLM is not.

3. What to report

Specificity (did neighbors survive?), generalization (paraphrase, multilingual), and a retain metric. If your project needs “remove this,” say which operational definition you used (extractive QA accuracy, membership inference, verbatim string match). Verbatim drop can hide latent knowledge.

4. Practice

1. ROME writes a rank-one bump in one MLP. Why might a paraphrase still produce the old fact?
2. You unlearn a 200-document set and only measure those 200 prompts. What did you fail to measure?
3. Editing `Paris → Berlin` is a key rewrite. Unlearning “all EU capitals” is a set. Which one should you *not* expect ROME’s single write to do well?